

MAKING TEXT EMBEDDERS FEW-SHOT LEARNERS

Chaofan Li^{1,2,*}, Minghao Qin^{1,3,*}, Shitao Xiao¹, Jianlyu Chen^{1,4}, Kun Luo^{1,3}, Yingxia Shao²,
Defu Lian⁴, Zheng Liu^{1†}

1: Beijing Academy of Artificial Intelligence

2: Beijing University of Posts and Telecommunications

3: Chinese Academy of Sciences

4: University of Science and Technology of China

{cfli, shaoyx}@bupt.edu.cn qinminghao24@ia.ac.cn

stxiao@baai.ac.cn chenjianlv@mail.ustc.edu.cn

liandefu@ustc.edu.cn {luokun695, zhengliu1026}@gmail.com

ABSTRACT

Large language models (LLMs) with decoder-only architectures demonstrate remarkable in-context learning (ICL) capabilities. This feature enables them to effectively handle both familiar and novel tasks by utilizing examples provided within their input context. Recognizing the potential of this capability, we propose leveraging the ICL feature in LLMs to enhance the process of text embedding generation. To this end, we introduce a novel model **bge-en-icl**, which employs few-shot examples to produce high-quality text embeddings. Our approach integrates task-related examples directly into the query side, resulting in significant improvements across various tasks. Additionally, we have investigated how to effectively utilize LLMs as embedding models, including various attention mechanisms, pooling methods, etc. Our findings suggest that retaining the original framework often yields the best results, underscoring that simplicity is best. Experimental results on the MTEB and AIR-Bench benchmarks demonstrate that our approach sets new state-of-the-art (SOTA) performance. Our model, code and dataset are freely available at <https://github.com/FlagOpen/FlagEmbedding>.

1 INTRODUCTION

Text embeddings are vector representations that capture the semantic and contextual meaning of natural language text. They play a pivotal role in natural language processing (NLP) tasks, facilitating a wide range of applications such as information retrieval, text classification, item recommendation, and question answering (Karpukhin et al., 2020; Xiong et al., 2020; Lu et al., 2020). Pre-trained bidirectional encoder and encoder-decoder architectures have been widely adopted as backbone models for embedding model, owing to their effectiveness in producing high-quality vector embeddings for text thanks to their extensive pre-training (Xiao et al., 2022; Gao et al., 2021).

Recent advancements in LLMs have significantly shifted the focus towards embedding models that rely primarily on decoder-only architectures (Ma et al., 2023; Li et al., 2024; Wang et al., 2023). These LLM-based embedding models have demonstrated remarkable improvements in in-domain accuracy and generalization, particularly when trained using supervised learning approaches (Wang et al., 2023). However, despite these advances, embedding models still struggle to follow unseen task instructions and execute complex retrieval tasks Su et al. (2024); Weller et al. (2024). This limitation stems from a mismatch between the relatively narrow range of instructions encountered during training and the broader variety of real-world text embedding tasks.

In-context learning (ICL) is a core capability of LLMs, enabling them to incorporate task-specific examples directly into input prompts to generate desired outputs (Radford et al., 2019; Brown, 2020;

*Co-first authors

†Corresponding author

Gao et al., 2020). The scope of ICL extends beyond tasks seen during training; it enables LLMs to generalize to new and complex tasks by learning patterns from the provided examples. This allows LLMs to adapt dynamically to novel tasks without additional training, making them highly applicable to large-scale, real-world scenarios (Wei et al., 2022; Yao et al., 2022; Dong et al., 2022).

Recognizing the robust ICL abilities of LLMs, in this study, we propose to generate more adaptable text embeddings with ICL strategy. Specifically, we guide the model by including task-specific examples directly within the query prompt. By doing so, we leverage the ICL capabilities of LLMs to produce embeddings that are not only more relevant to the specific domain but also more generalizable across various contexts.

Moreover, LLMs are predominantly utilized for text generation tasks, and adapting them for embedding representation tasks requires specific fine-tuning strategies. Recent studies have introduced various approaches, including the generation of high-quality training data through LLMs (Wang et al., 2023), modifications to attention mechanisms, and changes in pooling methods (Ma et al., 2023; Li et al., 2024). Following previous works (Muennighoff et al., 2024; BehnamGhader et al., 2024), we investigate how to effectively utilize LLMs as embedding models by modifying various architectures, e.g., bidirectional attention, meaning pooling, etc. Our experimental findings indicate that in the ICL scenario, making complex modifications to the models does not lead to significant improvements. Surprisingly, the best results are obtained using the original, unmodified architecture. By employing only the ICL strategy, our model **bge-en-icl** achieves state-of-the-art (SOTA) results on both the MTEB and AIR-Bench benchmarks. We have also released a multi-language embedding model **bge-multilingual-gemma2** and a lightweight reranker **bge-reranker-v2.5-gemma2-lightweight**. The lightweight reranker also serves as the teacher model for training embedding models through distillation. Further details are provided in Appendices C and D.

In summary, the key contributions of our work are as follows:

- We propose **bge-en-icl**, which incorporate few-shot examples into the query side to enhance the query embeddings. This integration leverages the in-context learning (ICL) capabilities of large language models (LLMs) in text embedding tasks.
- We rethink and explore how to effectively utilize LLMs as embedding models by evaluating various attention mechanisms, pooling methods, and the incorporation of passage prompts. Our findings highlight that simplicity is best; simply combining ICL capabilities with embedding models can achieve excellent performance.
- In contrast to other leading models on the MTEB benchmark, we provide open access to our model checkpoint, dataset, and training scripts.

2 RELATED WORK

Text embedding is a critical research direction in the field of information retrieval, with wide-ranging applications including web search, question answering, and dialogue systems. The fundamental principle involves encoding both queries and documents into embedding vectors within the same latent space. By calculating similarity scores between these vectors, effective retrieval is achieved. In recent years, numerous studies have leveraged pre-trained language models such as BERT (Devlin, 2018), T5 (Raffel et al., 2020), and RoBERTa (Liu, 2019) as the backbone for embedding models. These models have consistently demonstrated superior performance compared to traditional sparse retrieval methods.

The capability of the backbone is a crucial determinant in the effectiveness of retrieval systems. (Luo et al., 2024) have demonstrated that performance improves with increased scale and extensive pre-training. Currently, numerous studies have explored the effectiveness of utilizing LLMs as backbone encoders for text embedding tasks.

Repllama (Ma et al., 2023) fine-tuned Llama-2 to serve as both a dense retriever and a reranker, demonstrating the effectiveness of applying large language models (LLMs) in text embedding tasks. To further align LLMs with text embedding tasks, Llama2Vec (Li et al., 2024) introduced two pre-training tasks specifically designed to enhance the model’s performance in such tasks, which led to significant improvements on the BEIR benchmark. E5-mistral and Gecko (Wang et al., 2023; Lee et al., 2024b) advanced the training of LLM-based embedding models through the use of syn-

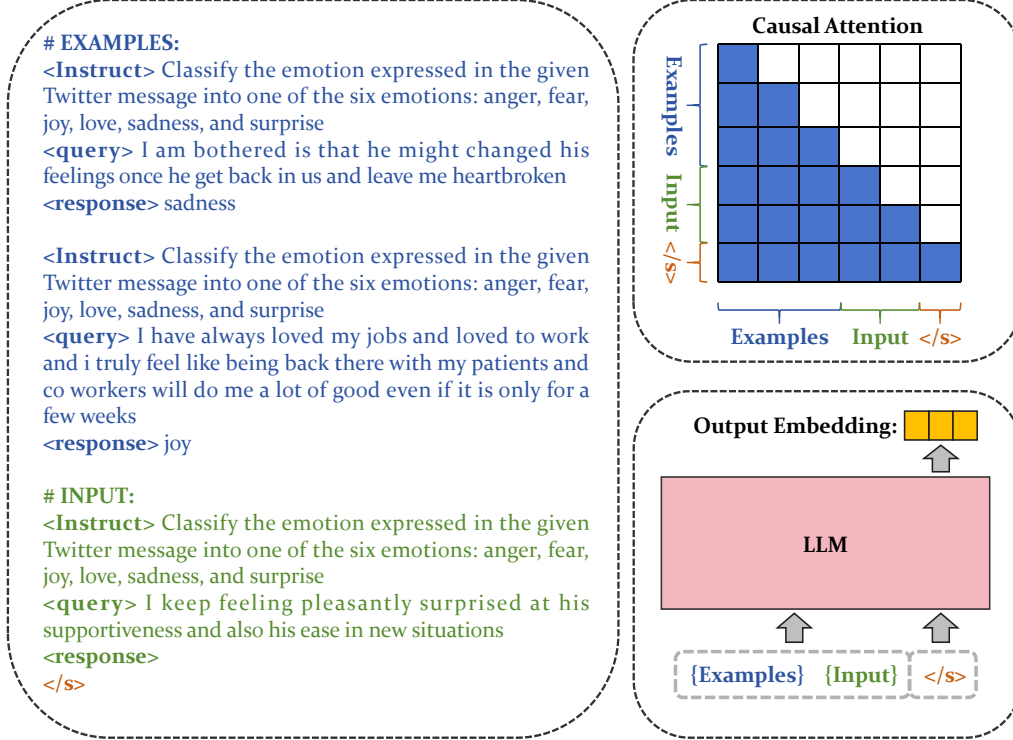


Figure 1: The architecture of the ICL-based model.

thetic data, markedly boosting their performance across a diverse range of retrieval and non-retrieval tasks. NV-Embed (Lee et al., 2024a) innovatively proposed a latent attention layer to replace conventional pooling methods and implemented a two-stage training strategy to address the challenge of false negatives in non-retrieval tasks. This model has shown strong performance in both retrieval and non-retrieval domains. Additionally, GRIT (Muennighoff et al., 2024) successfully integrated text embedding and generation within a single LLM, achieving performance levels on par with specialized models focused solely on either embedding or generation. In the exploration of LLMs as embedding models from an unsupervised perspective, LLM2Vec (BehnamGhader et al., 2024) presented a novel unsupervised method to transform decoder-only LLMs into embedding models. This approach demonstrated significant potential for modifying LLM backbone encoders to perform retrieval without any supervision. Similarly, PromptReps (Zhuang et al., 2024) leveraged chat-based LLMs aligned with human preferences to generate high-quality dense representations in an unsupervised manner.

The LLM-based embedding models mentioned above exhibit commendable performance across both retrieval and non-retrieval tasks. However, much of the existing work has disproportionately focused on altering model architectures, thereby neglecting the intrinsic capabilities of LLMs. Even models like GritLM, which integrate generation and embedding functionalities, fail to fully exploit the potential ICL capabilities of LLMs within the embedding process. By leveraging the innate ICL capabilities of LLMs, embedding models can be more versatile and adapt to diverse scenarios without necessitating additional fine-tuning. Our model not only achieves SOTA results on the MTEB and AIR-Bench benchmarks but also effectively utilizes the inherent strengths of LLMs across tasks.

3 METHODOLOGY

3.1 IN-CONTEXT LEARNING FOR EMBEDDING MODELS

Previous embedding models often involve directly inputting the query into the model to generate target embeddings. However, this method struggles to handle tasks with different intents, limiting the model’s adaptability and generalization capabilities. To address this, researchers have introduced task instructions (Su et al., 2022) appended to queries, enabling a single embedding model to generalize across tasks in various domains by altering the instructions.

Despite these advances, studies such as Su et al. (2024); Weller et al. (2024) reveal that embedding models have a limited ability to follow unseen embedding task instructions and conduct complex retrieval tasks. This limitation arises from a gap between the limited diversity of instructions seen during training and the vast range of real-world scenarios. Inspired by the ability of LLMs to generalize to unseen tasks through in-context learning (ICL), we explore whether embedding models can be enhanced by leveraging ICL, thereby significantly improving their generalization and versatility across diverse embedding tasks with various user intents.

In this work, we demonstrate the potential of embedding models to benefit from ICL through few-shot contrastive training. Consider a query-passage pair (q_i, p_i) in an embedding task. We first construct an example template as follows:

$$\langle \text{Instruct} \rangle \{ \text{task_definition} \} \langle \text{query} \rangle \{ q_i \} \langle \text{response} \rangle \{ p_i \} \quad (1)$$

Here, "*task_definition*" represents the description of the specific embedding task. This example template is applied to new input queries for each embedding task (Figure 1). For a relevant query-passage pair (q^+, p^+) , the modified query q_{exp}^+ is constructed as follows:

$$\{ \text{example 1} \} \dots \{ \text{example n} \} \langle \text{Instruct} \rangle \{ \text{task_definition} \} \langle \text{query} \rangle \{ q^+ \} \langle \text{response} \rangle \quad (2)$$

All modified queries and passages in the corpus are encoded using the same LLM to obtain their embedding representations. Specifically, we append an [EOS] token to the end of the input modified queries and passages, feeding them into the LLM to obtain embeddings $(h_{q_{\text{exp}}^+}, h_{p^+})$ by extracting the final layer's [EOS] vector. We employ the standard InfoNCE (Izacard et al., 2021) loss function L , utilizing both in-batch negatives and hard negatives for training:

$$L = -\log \frac{\exp(s(q_{\text{exp}}^+, p_i^+))}{\exp(s(q_{\text{exp}}^+, p_i^+)) + \sum_j \exp(s(q_{\text{exp}}^+, p_j^-))} \quad (3)$$

p_j^- denotes the set of negative passages, and $s(q, p)$ is the scoring function between the query and passage. In this work, we adopt a temperature-scaled cosine similarity function defined as:

$$s(q, p) = \frac{1}{\tau} \cos(h_q, h_p) \quad (4)$$

where τ is a temperature hyperparameter, which is fixed at 0.02 during training.

3.2 REPRESENTATION METHOD

The attention mechanism in LLM-based embedding models is typically unidirectional, aligned with the next-token prediction task fundamental to their pre-training (Touvron et al., 2023).

However, recent studies indicate that unidirectional attention may limit the model's capacity for representation learning. Evidence suggests that bidirectional attention is more effective at integrating contextual information, resulting in improved performance on certain tasks. For example, LLM2Vec (BehnamGhader et al., 2024) introduces an additional training phase with a masked token prediction task, preconditioning the model for bidirectional attention. Approaches such as NV-Embed (Lee et al., 2024a) and GritLM (Muennighoff et al., 2024) replace unidirectional attention with bidirectional attention during the embedding training phase, often employing mean pooling or more sophisticated latent attention layers to obtain representations for queries and passages.

Despite these advances, we argue that incorporating bidirectional attention during embedding fine-tuning creates a mismatch with the model's pre-training design, potentially undermining its in-context learning and generative properties. To address the trade-off between enhancing embedding representations for specific tasks and preserving the model's inherent generative properties for deep semantic pattern understanding, our approach retains the unidirectional attention mechanism, consistent with the majority of existing embedding methods.

We use the [EOS] token's output embedding as the vector representation for queries and passages, positioning it at the end of inputs to capture both semantic and ICL patterns through causal attention

mechanisms, thereby aligning with the foundational pretraining methodology of LLMs. Specifically, given the tokenized input sequence T : $[\text{BOS}], t_1, \dots, t_N$ is sent into the LLM (Figure 1):

$$h_t = \text{LLM}(T)[\text{EOS}] \quad (5)$$

The text embedding is taken from the output embedding of the special token $[\text{EOS}]$.

3.3 ICL-BASED INSTRUCTION-TUNING

While previous works (Wang et al., 2023; Lee et al., 2024a) have proposed the training method of instruction-tuning, which incorporates a large number of task-specific instructions during the training process, enabling the model to adapt to various downstream retrieval tasks based on different instructions, it is not applicable to the ICL strategy. As demonstrated by GRIT (Muennighoff et al., 2024), directly supplying few-shot examples when generating embeddings can actually degrade model performance.

To incorporate ICL capabilities into models, we need to modify the conventional instruction tuning strategy. Our approach involves integrating ICL abilities during the training phase. Specifically, we provide task-relevant examples to the query throughout the training process, allowing the model to develop ICL capabilities as it learns.

Recognizing the risk of compromising zero-shot capabilities if examples are consistently provided during training, we propose a dynamic training process. In each training step, queries are supplied with a variable number of few-shot examples, ranging from zero to n , determined by a sampling function. This approach maintains a balance between developing ICL abilities and preserving zero-shot performance.

To further enhance the model’s ICL capabilities, we introduce an innovative technique for examples selection. By incorporating in-batch pairs as few-shot examples, we train the model to better differentiate between examples and inputs, aims to improve the model’s ability to generate reliable embeddings based on the provided examples.

4 EXPERIMENTENS

In this section, we examine the effectiveness of the ICL training pipeline and reconsider the training methodologies for LLM-based embedding models.

- **RQ 1:** What is the effectiveness of the ICL training strategy for both zero-shot and few-shot learning scenarios?
- **RQ 2:** How does the ICL training strategy impact performance compared to traditional training methods?
- **RQ 3:** How does the integration of in-batch examples affect the performance of the ICL training strategy.
- **RQ 4:** What are the implications of replacing a causal attention mask with a bidirectional attention mask within the framework of LLMs?
- **RQ 5:** What is the impact of various representation strategies, including last token pooling and mean pooling, on model performance?
- **RQ 6:** Do passage-based prompts enhance performance in the ICL training strategy?

4.1 SETUP

LLM. Following E5-Mistral (Wang et al., 2023), SFR, and NV-Embedder (Lee et al., 2024a), we have adopted Mistral-7B (Jiang et al., 2023) as the backbone for our framework.

Evaluation. We evaluate the performance of our model on MTEB (Muennighoff et al., 2022) and AIR-Bench. MTEB is a comprehensive benchmark designed to evaluate the performance of text embedding models. AIR-Bench is dedicated to the evaluation of retrieval performance, its testing data is automatically generated by large language models without human intervention.

Training Data. To ensure a fair comparison, we use the same public datasets from E5-Mistral (Wang et al., 2023), which includes ELI5 (Fan et al., 2019), HotpotQA (Yang et al., 2018), FEVER (Thorne et al., 2018), MIRACL (Zhang et al., 2023), MSMARCO passage and document ranking (Nguyen et al., 2016), NQ (Karpukhin et al., 2020), NLI (Gao et al., 2021), SQuAD (Karpukhin et al., 2020), TriviaQA (Karpukhin et al., 2020), Quora Duplicate Questions (DataCanary et al., 2017), MrTyDi (Zhang et al., 2021), DuReader (Qiu et al., 2022), and T2Ranking (Xie et al., 2023), all of which are also used for LLM2Vec (BehnamGhader et al., 2024).

However, methods that typically perform exceptionally well, such as NV-Embedder (Lee et al., 2024a) and SFR, often require more training data. Additionally, some of these methods, such as GTE-Qwen2 (Li et al., 2023), do not disclose their sources of training data. In response, we have developed an enhanced version of our model that leverages a more comprehensive dataset, which includes the following training sets:

- **Retrieval:** ELI5, HotpotQA, FEVER, MSMARCO passage and document ranking, NQ, NLI, SQuAD, TriviaQA, Quora Duplicate Questions, Arguana (Wachsmuth et al., 2018), and FiQA (Maia et al., 2018).
- **Reranking:** SciDocsRR (Cohan et al., 2020) and StackOverFlowDupQuestions (Liu et al., 2018).
- **Classification:** AmazonReviews-Classification (McAuley & Leskovec, 2013), AmazonCounterfactual-Classification (O’Neill et al., 2021), Banking77-Classification (Casanueva et al., 2020), Emotion-Classification (Saravia et al., 2018), TweetSentimentExtraction-Classification (Maggie, 2020), MTOPIntent-Classification (Li et al., 2020), IMDB-Classification (Maas et al., 2011), ToxicConversations-Classification (Adams et al., 2019).
- **Clustering:** {Arxiv/Biorxiv/Medrxiv/Reddit/StackExchange}-Clustering-{S2S/P2P}, TwentyNewsgroups-Clustering (Lang, 1995).
- **STS:** STS12 (Agirre et al., 2012), STS22 (Chen et al., 2022), STS-Benchmark (Cer et al., 2017).

Training Detail. We fine-tune the Mistral-7B model using a contrastive loss and conduct the process over a single epoch. For efficient fine-tuning, we employ Low-Rank Adaptation (LoRA) (Hu et al., 2021), setting the LoRA rank to 64 and the LoRA alpha to 32, with a learning rate of $1e-4$. For retrieval tasks, we use in-batch negatives, a strategy not adopted for other tasks. Each dataset incorporates 7 hard negatives. The batch size is set to 512 for retrieval tasks and 256 for other types of tasks. We maintain consistency by using the same dataset throughout one training step, and the maximum sequence length is set at 512 tokens. To distill the score from reranker in retrieval tasks, we use the bge-reranker model as the teacher. For in-context learning training, we implement a randomized sampling method. For each query, we select between 0 to 5 examples from the in-batch training data. The maximum allowable lengths for example queries and documents are set to 256 tokens each, and the combined length for a query with examples is set at 2048 tokens.

Evaluation. We evaluate the performance of our model under both zero-shot and few-shot conditions. In the few-shot scenario, a consistent set of in-context examples is applied to each query. The examples utilized for evaluation are sourced from training datasets. In cases where training datasets are unavailable, examples are generated using ChatGPT.

4.2 MAIN RESULTS

MTEB. Table 1 presents the performance of our model, *bge-en-icl*, evaluated on the MTEB benchmark. This evaluation contrasts the results obtained from using the full dataset with those obtained from using only the public dataset. When leveraging the full dataset, our model demonstrates strong capabilities in both zero-shot and few-shot settings, achieving SOTA results in few-shot scenarios. However, it is important to note that the use of full datasets may introduce inconsistencies, as different models often rely on varying datasets. Notably, many of these models do not disclose the specific datasets they use, leading to potential unfair comparisons.

For a fairer comparison and to better understand the impact of in-context learning, we conduct an evaluation using only the public dataset. Under these constraints, our model’s performance in

Task # of datasets →	Retr. 15	Rerank. 4	Clust. 11	PairClass. 3	Class. 12	STS 10	Summ. 1	Avg. 56
w/ full data								
E5-mistral-7b-instruct	56.90	60.21	50.26	88.34	78.47	84.66	31.40	66.63
GritLM-7B	57.41	60.49	50.61	87.16	79.46	83.35	30.37	66.76
SFR-Embedding	59.00	60.64	51.67	88.54	78.33	85.05	31.16	67.56
Linq-Embed-Mistral	60.19	60.29	51.42	88.35	80.20	84.97	30.98	68.17
voyage-large-2-instruct	58.28	60.09	53.35	89.24	81.49	84.31	30.84	68.23
NV-Embed-v1	59.36	60.59	52.80	86.91	87.35	82.84	31.20	69.32
bge-multilingual-gemma2	59.24	59.72	54.65	85.84	88.08	83.88	31.20	69.88
stella_en_400M_v5	58.97	60.16	56.70	87.74	86.67	84.22	31.66	70.11
gte-Qwen2-7B-instruct	60.25	61.42	56.92	85.79	86.58	83.04	31.35	70.24
SFR-Embedding-2_R	60.18	60.14	56.17	88.07	89.05	81.26	30.71	70.31
stella_en_1.5B_v5	61.01	61.21	57.69	88.07	87.63	84.51	31.49	71.19
bge-en-icl (zero-shot)	61.67	59.66	57.51	86.93	88.62	83.74	30.75	71.24
bge-en-icl (few-shot)	62.16	59.82	57.89	88.14	88.95	84.24	30.77	71.67
w/ public data only								
E5-mistral-7b-instruct	52.78	60.38	47.78	88.47	76.80	83.77	31.90	64.56
GritLM-7B	53.10	61.30	48.90	86.90	77.00	82.80	29.40	64.70
LLM2Vec-Mistral-supervised	55.99	58.42	45.54	87.99	76.63	84.09	29.96	64.80
bge-en-icl (zero-shot)	59.59	56.85	42.61	87.87	75.47	83.30	29.52	64.67
bge-en-icl (few-shot)	60.08	56.67	46.55	88.51	77.31	83.69	30.68	66.08

Table 1: Top MTEB leaderboard models as of August 27, 2024.

Domain # of datasets →	wiki 1	web 1	news 1	healthcare 1	law 1	finance 1	arxiv 1	msmarco 1	Avg. 8
w/ full data									
E5-mistral-7b-instruct	61.67	44.41	48.18	56.32	19.32	54.79	44.78	59.03	48.56
SFR-Embedding	63.46	51.27	52.21	58.76	23.27	56.94	47.75	58.99	51.58
NV-Embed-v1	62.84	50.42	51.46	58.53	20.65	49.89	46.10	60.27	50.02
Linq-Embed-Mistral	61.04	48.41	49.44	60.18	20.34	50.04	47.56	60.50	49.69
gte-Qwen2-7B-instruct	63.46	51.20	54.07	54.20	22.31	58.20	40.27	58.39	50.26
stella_en_1.5B_v5	61.99	50.88	53.87	58.81	23.22	57.26	44.81	61.38	51.53
bge-en-icl (zero-shot)	64.61	54.40	55.11	57.25	25.10	54.81	48.46	63.71	52.93
bge-en-icl (few-shot)	64.94	55.11	56.02	58.85	28.29	57.16	50.04	64.50	54.36
w/ public data only									
bge-en-icl (zero-shot)	64.82	54.96	55.82	57.06	28.87	54.46	49.60	63.25	53.60
bge-en-icl (few-shot)	66.98	56.38	57.17	59.54	32.03	58.81	51.36	65.05	55.92

Table 2: QA (en, nDCG@10) performance on AIR-Bench 24.04.

the zero-shot scenario is on par with, or slightly below, that of other models such as LLM2Vec and GritLM. However, in the few-shot settings, our model show significant enhancements ($\uparrow 1.41$), particularly in the classification and clustering tasks that were not part of the training data. These improvements underscore the potential advantages of in-context learning, emphasizing its efficacy in adapting to tasks beyond the direct scope of initial training parameters. Furthermore, in contrast to training exclusively with public datasets, the utilization of full training data effectively familiarizes the model with these datasets. As a result, the model’s ability to generalize effectively is compromised, leading to only a modest improvement in few-shot settings ($\uparrow 0.43$).

AIR-Bench. The performance of our model is also evaluated using the AIR-Bench dataset. As illustrated in Tables 2 and 3, the model demonstrates superior performance compared to prior models in both zero-shot and few-shot scenarios, excelling across qa and long-doc tasks. Notably, there is no overlap between the training dataset and the evaluation data for these tasks, highlighting the robustness of the model in scenarios with limited prior exposure. In the few-shot setting, the model exhibits significant improvements over the zero-shot scenario, achieving gains of 1.43 points in the qa task and 1.08 points in the long-doc task. This improvement underscores the efficacy of in-context learning in enhancing the model’s generalization capabilities.

Domain # of datasets →	arxiv 4	book 2	healthcare 5	law 4	Avg. 15
w/ full data					
text-embedding-3-large	74.53	73.16	65.83	64.47	68.77
E5-mistral-7b-instruct	72.14	72.44	68.44	62.92	68.49
SFR-Embedding	72.79	72.41	67.94	64.83	69.00
NV-Embed-v1	77.65	75.49	72.38	69.55	73.45
Linq-Embed-Mistral	75.46	73.81	71.58	68.58	72.11
gte-Qwen2-7B-instruct	63.93	68.51	65.59	65.26	65.45
stella_en_1.5B_v5	73.17	74.38	70.02	69.32	71.25
bge-multilingual-gemma2	71.77	76.46	73.96	70.86	72.88
bge-en-icl (zero-shot)	78.30	78.21	73.65	67.09	73.75
bge-en-icl (few-shot)	79.63	79.36	74.80	67.79	74.83
w/ public data only					
bge-en-icl (zero-shot)	79.73	78.66	72.88	70.59	74.86
bge-en-icl (few-shot)	79.82	80.37	74.60	71.66	75.98

Table 3: Long-Doc (en, Recall@10) performance on AIR-Bench 24.04.

However, when the model is trained exclusively using public data, it achieves better results compared to training with the full dataset. This could be attributed to the presence of an excessive amount of MTEB-related data, such as clustering and classification, within the full dataset. Such data might introduce the risk of overfitting, thereby potentially hampering the model’s generalization performance on the AIR-Bench dataset.

4.3 IN-CONTEXT LEARNING

Task # of datasets →	Retr. 15	Rerank. 4	Clust. 11	PairClass. 3	Class. 12	STS 10	Summ. 1	Avg. 56
w/ full data								
w/o in-context learning	59.11	57.02	42.60	87.99	76.27	83.93	30.50	64.83
w/ fix examples (zero-shot)	48.98	56.48	41.84	85.94	74.38	84.31	29.68	61.50
w/ fix examples (few-shot)	59.00	56.90	45.75	88.54	75.56	84.67	30.66	65.46
w/ in-batch examples (zero-shot)	59.59	56.85	42.61	87.87	75.47	83.30	29.52	64.67
w/ in-batch examples (few-shot)	60.08	56.67	46.55	88.51	77.31	83.69	30.68	66.08

Table 4: Evaluation of various ICL strategies on the MTEB Benchmark.

To evaluate the impact of the ICL strategy, we conduct a series of ablation studies using the MTEB benchmark. In these studies, we compare the performance of models fine-tuned with the ICL strategy against those fine-tuned without it. Specifically, for ICL training, we employ two distinct training approaches: fixed examples and in-batch examples. In the fixed examples approach, each task was trained using three predetermined examples.

In Table 4, we present various results from our experiment. When the model is trained without ICL strategy, its average performance is 64.83. This performance is comparable to GritLM (Muennighoff et al., 2024), LLM2Vec (BehnamGhader et al., 2024), etc. When fixed examples are used during ICL training, there is a significant decline in zero-shot evaluation performance, with a decrease of 3.33 points. This decline is attributed to the model’s consistent exposure to specific training examples, which can impair its zero-shot capabilities. On the other hand, in few-shot scenarios, the model demonstrates improved performance, exceeding zero-shot results by 3.96 points and surpassing models trained without ICL by 0.63 points. This also confirms the effectiveness of the ICL strategy.

Meanwhile, the use of in-batch examples, where training may involve zero examples, has preserved the zero-shot capability of the model. There is a modest decrease of 0.16 points compared to the model trained without ICL. Notably, in few-shot scenarios, the performance of the model employing in-batch examples rises to 66.08 ($\uparrow 1.25$), indicating a robust improvement. Furthermore, when compared with the model utilizing fixed examples, the model trained with in-batch examples displays

superior performance in tasks that diverge significantly from the training data, such as classification and clustering tasks.

4.4 ATTENTION

Task # of datasets →	Retr. 15	Rerank. 4	Clust. 11	PairClass. 3	Class. 12	STS 10	Summ. 1	Avg. 56
causal attention & last token pooling								
w/o in-context learning	59.11	57.02	42.60	87.99	76.27	83.93	30.50	64.83
w/ in-context learning (zero-shot)	59.59	56.85	42.61	87.87	75.47	83.30	29.52	64.67
w/ in-context learning (few-shot)	60.08	56.67	46.55	88.51	77.31	83.69	30.68	66.08
causal attention & mean pooling								
w/o in-context learning	58.50	53.74	36.82	82.14	72.37	77.62	29.10	61.03
bidirectional attention & last token pooling								
w/o in-context learning	59.59	56.96	44.34	87.61	74.77	83.81	30.12	64.96
w/ in-context learning (zero-shot)	59.77	58.09	44.04	87.87	75.35	83.97	29.75	65.19
w/ in-context learning (few-shot)	60.23	57.81	44.45	88.64	77.00	83.77	29.99	65.74
bidirectional attention & mean pooling								
w/o in-context learning	59.13	57.03	43.44	87.25	75.03	84.08	29.17	64.73
w/ in-context learning (zero-shot)	59.53	57.48	43.88	88.12	74.86	83.64	29.58	64.90
w/ in-context learning (few-shot)	59.42	57.29	44.93	88.36	75.26	83.75	29.60	65.18

Table 5: Results of different attention and pooling mechanisms on the MTEB Benchmark.

Recent studies have explored modifying causal attention in LLMs to adopt bidirectional attention and employ mean pooling for embedding generation. Notably, models such as GritLM (Muenighoff et al., 2024), NV-Embed (Lee et al., 2024a), and LLM2Vec (BehnamGhader et al., 2024) have utilized these techniques with considerable experimental success. Motivated by these advancements, we explore the potential benefits of implementing bidirectional attention in the ICL scenario. Specifically, we investigate the impacts of various attention and pooling mechanisms, including causal and bidirectional attention, coupled with last token pooling and mean pooling. In a causal attention framework, each token is limited to accessing only preceding tokens’ information and not the subsequent ones. Consequently, employing mean pooling tends to yield suboptimal results because of this restriction. We find that the model could not be trained effectively under the ICL setting. Therefore, only results from experiments without ICL are presented in this specific configuration.

Table 5 presents our experimental setup and results in both non-ICL and ICL scenarios. It demonstrates that in non-ICL scenarios, most methods yield consistent performance, aside from the combination of causal attention with mean pooling. In contrast, within ICL scenarios, the integration of causal attention and last token pooling emerges as the superior approach. This configuration appears to resonate with the competencies fostered during the initial training phase of the model, suggesting a strong alignment with the foundational strategies employed during pre-training. Moreover, shifting from causal attention to bidirectional attention does not result in significant improvements, and mean pooling is not necessary for the implementation of bidirectional attention.

Additionally, configurations utilizing bidirectional attention paired with last token pooling are notably effective, excelling in both non-ICL and zero-shot scenarios. This configuration’s performance is also pronounced in few-shot reranking tasks, highlighting its adaptability and potential applicability across various demands.

4.5 PROMPT

Recently, most LLM-based embedding models have incorporated instruction-based prompts on the query side. However, there has been limited investigation into the efficacy of utilizing prompts on the passage side. To address this gap, our study introduces and explores the use of prompts on the passage side. The specific prompt employed in our study is as follows:

$$\{\text{passage}\} \text{ Summarize the above passage:} \quad (6)$$

Table 6 presents the results obtained using passage prompts. The results demonstrate that the integration of passage prompts leads to a significant decline in performance across all tasks except

Task # of datasets →	Retr. 15	Rerank. 4	Clust. 11	PairClass. 3	Class. 12	STS 10	Summ. 1	Avg. 56
w/o passage prompt (zero-shot)	59.59	56.85	42.61	87.87	75.47	83.30	29.52	64.67
w/o passage prompt (few-shot)	60.08	56.67	46.55	88.51	77.31	83.69	30.68	66.08
w/ passage prompt (zero-shot)	59.50	46.84	39.57	81.25	71.41	80.38	30.26	61.61
w/ passage prompt (few-shot)	59.93	46.39	39.40	82.25	72.00	79.81	30.97	61.74

Table 6: Comparative results of different prompts on the MTEB benchmark.

retrieval. This indicates that further exploration and experimentation are needed when employing prompts at the passage level.

5 CONCLUSION

In this paper, we explore the utilization of in-context learning (ICL) derived from large language models (LLMs) for generating text embeddings and investigate various methods of LLMs as embedding models. Specifically, we examine the integration of attention mechanisms, different pooling methods, and passage prompts. We advocate for maintaining the model’s original architecture while embedding in-context learning capabilities into the dense retrieval process. Our approach necessitates no modifications to the model’s architecture; instead, it involves altering the prompt on the query side to include in-context learning features in the embedding generation task. Despite its simplicity, our method proves highly effective on the MTEB and AIR-Bench benchmarks.

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A INSTRUCTION

Task Name	Instruction Template
ArguAna	Given a claim, find documents that refute the claim.
ClimateFEVER	Given a claim about climate change, retrieve documents that support or refute the claim.
CQADupStack	Given a question, retrieve detailed question descriptions from Stackexchange that are duplicates to the given question.
DBPedia	Given a query, retrieve relevant entity descriptions from DBPedia.
FEVER	Given a claim, retrieve documents that support or refute the claim.
FiQA2018	Given a financial question, retrieve user replies that best answer the question.
HotpotQA	Given a multi-hop question, retrieve documents that can help answer the question.
MSMARCO	Given a web search query, retrieve relevant passages that answer the query.
NFCorpus	Given a question, retrieve relevant documents that best answer the question.
Natural Question	Given a question, retrieve Wikipedia passages that answer the question.
QuoraRetrieval	Given a question, retrieve questions that are semantically equivalent to the given question.
SCIDOCS	Given a scientific paper title, retrieve paper abstracts that are cited by the given paper.
SciFact	Given a scientific claim, retrieve documents that support or refute the claim.
Touche2020	Given a question, retrieve detailed and persuasive arguments that answer the question.
TREC-COVID	Given a query, retrieve documents that answer the query.
STS*	Retrieve semantically similar text.
SummEval	Given a news summary, retrieve other semantically similar summaries.
AmazonCounterfactualClassification	Classify a given Amazon customer review text as either counterfactual or not-counterfactual.
AmazonPolarityClassification	Classify Amazon reviews into positive or negative sentiment.
AmazonReviewsClassification	Classify the given Amazon review into its appropriate rating category.
Banking77Classification	Given an online banking query, find the corresponding intents.
EmotionClassification	Classify the emotion expressed in the given Twitter message into one of the six emotions: anger, fear, joy, love, sadness, and surprise.
ImdbClassification	Classify the sentiment expressed in the given movie review text from the IMDB dataset.
MassiveIntentClassification	Given a user utterance as query, find the user intents.
MassiveScenarioClassification	Given a user utterance as query, find the user scenarios.
MTOPDomainClassification	Classify the intent domain of the given utterance in task-oriented conversation.
MTOPIntentClassification	Classify the intent of the given utterance in task-oriented conversation.
ToxicConversationsClassification	Classify the given comments as either toxic or not toxic.
TweetSentimentExtractionClassification	Classify the sentiment of a given tweet as either positive, negative, or neutral.
ArxivClusteringP2P	Identify the main and secondary category of Arxiv papers based on the titles and abstracts.
ArxivClusteringS2S	Identify the main and secondary category of Arxiv papers based on the titles.
BiorxivClusteringP2P	Identify the main category of Biorxiv papers based on the titles and abstracts.
BiorxivClusteringS2S	Identify the main category of Biorxiv papers based on the titles.
MedrxivClusteringP2P	Identify the main category of Medrxiv papers based on the titles and abstracts.
MedrxivClusteringS2S	Identify the main category of Medrxiv papers based on the titles.
RedditClustering	Identify the topic or theme of Reddit posts based on the titles.
RedditClusteringP2P	Identify the topic or theme of Reddit posts based on the titles and posts.
StackExchangeClustering	Identify the topic or theme of StackExchange posts based on the titles.
StackExchangeClusteringP2P	Identify the topic or theme of StackExchange posts based on the given paragraphs.
TwentyNewsgroupsClustering	Identify the topic or theme of the given news articles.
AskUbuntuDupQuestions	Retrieve duplicate questions from AskUbuntu forum.
MindSmallReranking	Retrieve relevant news articles based on user browsing history.
SciDocsRR	Given a title of a scientific paper, retrieve the titles of other relevant papers.
StackOverflowDupQuestions	Retrieve duplicate questions from StackOverflow forum.
SprintDuplicateQuestions	Retrieve duplicate questions from Sprint forum.
TwitterSemEval2015	Retrieve tweets that are semantically similar to the given tweet.
TwitterURLCorpus	Retrieve tweets that are semantically similar to the given tweet.
AIR-Bench	Given a question, retrieve passages that answer the question.

Table 7: The instruction we used on the MTEB and AIR-Bench benchmarks.

B DETAILED MTEB RESULTS

Dataset	NV-Embed-v1	bge-multilingual-gemma2	gte-Qwen2-7B-instruct	SFR-Embedding-2_R	stella_en_1.5B_v5	bge-en-icl (zero-shot)	bge-en-icl (few-shot)
ArguAna	68.21	77.37	64.27	62.34	65.27	82.76	83.08
ClimateFEVER	34.72	39.37	45.88	34.43	46.11	45.35	45.43
CQADupStack	50.51	47.94	46.43	46.11	47.75	47.23	47.31
DBPEDIA	48.29	51.37	52.42	51.21	52.28	50.42	51.63
FEVER	87.77	90.38	95.11	92.16	94.83	91.96	92.83
FiQA2018	63.10	60.04	62.03	61.77	60.48	58.77	59.67
HotpotQA	79.92	83.26	73.08	81.36	76.67	84.98	85.14
MSMARCO	46.49	45.71	45.98	42.18	45.22	46.72	46.79
NFCorpus	38.04	38.11	40.60	41.34	42.00	40.69	41.85
Natural Question	71.22	71.45	67.00	73.96	71.80	73.85	73.88
QuoraRetrieval	89.21	90.04	90.09	89.58	90.03	91.02	90.95
SCIDOCS	20.19	26.93	28.91	24.87	26.64	25.25	25.26
SciFact	78.43	72.05	79.06	85.91	80.09	78.33	79.09
Touche2020	28.38	30.26	30.57	28.18	29.94	29.67	30.48
TREC-COVID	85.88	64.27	82.26	87.28	85.98	78.11	79.08
BIOSSES	85.59	85.74	81.37	87.60	83.11	86.35	86.47
SICK-R	82.80	82.66	79.28	77.01	82.89	83.87	83.87
STS12	76.22	77.71	79.55	75.67	80.09	77.73	78.14
STS13	86.30	87.45	88.83	82.40	89.68	85.98	86.59
STS14	82.09	83.48	83.87	79.93	85.07	82.34	82.83
STS15	87.24	87.63	88.54	85.82	89.39	87.35	87.77
STS16	84.77	86.70	86.49	84.50	87.15	86.54	87.04
STS17	87.42	91.18	88.73	88.93	91.35	91.25	91.25
STS22	69.85	69.02	66.88	67.10	68.10	68.08	70.07
STSBenchmark	86.14	87.25	86.85	83.60	88.23	87.92	88.42
SummEval	31.20	31.20	31.35	30.71	31.49	30.75	30.77
SprintDuplicateQuestions	95.94	90.94	92.82	97.62	96.04	95.06	97.23
TwitterSemEval2015	78.73	79.64	77.96	78.57	80.58	78.54	79.34
TwitterURLCorpus	86.05	86.95	86.59	88.03	87.58	87.19	87.84
AmazonCounterfactual	95.12	89.48	91.31	92.72	92.87	92.88	93.15
AmazonPolarity	97.14	96.90	97.50	97.31	97.16	96.86	96.98
AmazonReviews	55.47	61.60	62.56	61.04	59.36	61.28	61.46
Banking77	90.34	92.53	87.57	90.02	89.79	91.42	91.49
Emotion	91.71	92.97	79.45	93.37	84.29	93.31	93.36
Imdb	97.06	96.66	96.75	96.80	96.66	96.91	96.91
MassiveIntent	80.07	82.05	85.41	85.97	85.83	82.26	82.93
MassiveScenario	81.74	84.40	89.77	90.61	90.20	83.92	85.60
MTOPDomain	96.51	98.61	99.04	98.58	99.01	97.99	98.42
MTOPIntent	89.77	95.51	91.88	91.30	92.78	93.56	94.00
ToxicConversations	92.60	87.34	85.12	91.14	88.76	93.16	93.17
TweetSentimentExtraction	80.60	78.86	72.58	79.70	74.84	79.90	79.93
Arxiv-P2P	53.76	54.91	54.46	54.02	55.44	54.42	54.44
Arxiv-S2S	49.59	50.28	51.74	48.82	50.66	49.17	49.33
Biorxiv-P2P	48.15	52.64	50.09	50.76	50.68	52.32	53.05
Biorxiv-S2S	44.74	49.20	46.65	46.57	46.87	48.38	48.38
Medrxiv-P2P	39.24	45.81	46.23	46.66	46.87	46.13	45.86
Medrxiv-S2S	36.98	44.11	44.13	44.18	44.65	44.20	44.33
Reddit	63.20	56.03	73.55	62.92	72.86	71.20	72.33
Reddit-P2P	68.01	65.83	74.13	72.74	75.27	72.17	72.72
StackExchange	74.99	66.21	79.86	76.48	80.29	81.29	81.32
StackExchange-P2P	42.04	45.74	49.41	48.29	49.57	45.53	46.05
TwentyNewsgroups	60.13	70.44	53.91	66.42	61.43	68.51	68.98
AskUbuntuDupQuestions	67.50	64.59	67.58	66.71	67.33	64.80	65.15
MindSmallRerank	30.82	31.79	33.36	31.26	33.05	30.60	30.60
SciDocsRR	87.26	87.60	89.09	87.29	89.20	86.90	86.96
StackOverflowDupQuestions	56.58	54.90	55.66	55.32	55.25	56.32	56.71
MTEB Average (56)	69.32	69.88	70.24	70.31	71.19	71.24	71.67

Table 8: MTEB results with full data.

Dataset	bge-en-icl (zero-shot)	bge-en-icl (few-shot)
ArguAna	55.81	55.41
ClimateFEVER	45.17	45.14
CQADupStack	46.03	46.46
DBPEDIA	50.79	51.14
FEVER	91.96	92.42
FiQA2018	58.49	58.15
HotpotQA	84.34	84.68
MSMARCO	46.52	46.56
NFCorpus	40.16	40.96
Natural Question	73.56	74.01
QuoraRetrieval	90.79	90.89
SCIDOCS	20.56	20.87
SciFact	78.10	79.65
Touche2020	33.64	34.93
TREC-COVID	77.89	79.95
BIOSSES	86.80	87.49
SICK-R	83.83	83.69
STS12	77.80	78.39
STS13	84.90	85.62
STS14	82.53	82.62
STS15	88.33	88.52
STS16	86.14	86.44
STS17	91.65	91.79
STS22	63.79	64.83
STSBenchmark	87.27	87.52
SummEval	29.52	30.68
SprintDuplicateQuestions	94.79	96.09
TwitterSemEval2015	81.53	82.04
TwitterURLCorpus	87.30	87.39
AmazonCounterfactual	80.78	83.36
AmazonPolarity	88.57	92.69
AmazonReviews	47.55	49.85
Banking77	87.57	88.70
Emotion	54.29	54.24
Imdb	81.14	84.96
MassiveIntent	78.54	79.24
MassiveScenario	79.27	82.00
MTOPDomain	95.57	96.61
MTOPIntent	85.32	88.19
ToxicConversations	63.58	64.68
TweetSentimentExtraction	63.47	63.16
Arxiv-P2P	47.22	48.97
Arxiv-S2S	42.87	45.35
Biorxiv-P2P	33.17	38.37
Biorxiv-S2S	35.00	37.05
Medrxiv-P2P	28.74	30.24
Medrxiv-S2S	28.10	31.45
Reddit	53.83	59.14
Reddit-P2P	64.40	65.51
StackExchange	57.50	68.61
StackExchange-P2P	34.21	36.01
TwentyNewsgroups	43.65	51.40
AskUbuntuDupQuestions	63.71	62.96
MindSmallRerank	27.90	27.90
SciDocsRR	84.31	84.24
StackOverflowDupQuestions	51.48	51.56
MTEB Average (56)	64.67	66.08

Table 9: MTEB results with public data only.

C MULTILINGUAL EMBEDDING MODEL

Considering that the LLM-based multilingual embedding models are still relatively scarce, we further train a LLM-based multilingual embedding model, *bge-multilingual-gemma2*, on a diverse range of languages and tasks. It is noted that *bge-multilingual-gemma2* is just our initial attempt, and we have not explored the in-context learning (ICL) capabilities of *bge-multilingual-gemma2*. The exploration of ICL capabilities in the multilingual embedding models is probably a future research topic. However, in our experiment, the new multilingual embedding model has already achieved excellent performance on multiple embedding benchmarks, and especially led to new state-of-the-art results on several multilingual benchmarks.

C.1 SETUP

LLM. XLM-RoBERTa (Conneau et al., 2020) demonstrated that the larger vocabulary size were beneficial for improving the multilingual capability of language models. Therefore, we employ Gemma-2-9b (Team et al., 2024) as the backbone for the new multilingual embedding model, considering that the vocabulary size of Gemma-2-9b is 256K, which is larger than the vocabulary size of other LLMs, such as Qwen2 (Yang et al., 2024) or Llama 3 (Dubey et al., 2024).

Dataset. In addition to MTEB (Muennighoff et al., 2022) and AIR-Bench¹, we also evaluate the multilingual capability of *bge-multilingual-gemma2* on MIRACL (Zhang et al., 2023), FR-MTEB (Ciancone et al., 2024), PL-MTEB (Poświata et al., 2024) and C-MTEB (Xiao et al., 2024).

Training Data. For the English training data, we use most of the datasets used by *bge-en-icl*. For the Chinese training data, in addition to the datasets used by BGE-M3 (Chen et al., 2024), more datasets in retrieval, classification, and clustering tasks are included. For the multilingual training data, we still use the two multilingual datasets used by BGE-M3. The full training data used by *bge-multilingual-gemma2* includes:

- **English:** The English datasets (refer to Section 4.1) used by *bge-en-icl* are employed, except for the MSMARCO document ranking dataset.
- **Chinese:** The Chinese datasets used by BGE-M3 (Chen et al., 2024) are employed. The retrieval datasets including the three domain-specific datasets in Multi-CPR (Long et al., 2022), the classification datasets including AmazonReviews-Classification (McAuley & Leskovec, 2013) and MultilingualSentiment-Classification (Mollanorozzy et al., 2023), and the clustering datasets including CSL-Clustering- $\{S2S/P2P2\}$ (Yudong et al., 2022) are additionally employed.
- **Multilingual:** Two multilingual retrieval datasets including MIRACL (Zhang et al., 2023) and Mr.TyDi (Zhang et al., 2021) are employed.

Training Detail. We fine-tune the Gemma-2-9b model using a contrastive loss and conduct the process over a single epoch. For efficient fine-tuning, we employ Low-Rank Adaptation (LoRA) (Hu et al., 2021), setting the LoRA rank to 64 and the LoRA alpha to 32, with a learning rate of $1e-4$. We use in-batch negatives only for retrieval tasks, where each dataset incorporates 7 hard negatives. For the retrieval tasks and the other tasks, we set the batch size to 512 and 256, respectively. We consistently use the same dataset throughout one step, and the maximum sequence length remains capped at 512 tokens. Meanwhile, we use *bge-reranker* as the teacher to distill our model in retrieval tasks.

Evaluation. On the MTEB benchmark, the instructions used by *bge-multilingual-gemma2* are consistent with the instructions used by *bge-en-icl*, which are shown in Table 7. The instructions used by *bge-multilingual-gemma2* on the C-MTEB, PL-MTEB and FR-MTEB benchmarks are available in Table 16. On the MIRACL benchmark, we use the same instruction for all 18 languages: “Given a question, retrieve Wikipedia passages that answer the question.”. On the AIR-Bench benchmark, for the sake of simplicity, we also use the same instruction for all datasets: “Given a question, retrieve passages that answer the question.”.

¹<https://github.com/AIR-Bench/AIR-Bench>

Model	Avg.	ar	bn	en	es	fa	fi	fr	hi	id	ja	ko	ru	sw	te	th	zh	de	yo
BM25	31.9	39.5	48.2	26.7	7.7	28.7	45.8	11.5	35.0	29.7	31.2	37.1	25.6	35.1	38.3	49.1	17.5	12.0	56.1
mDPR	41.8	49.9	44.3	39.4	47.8	48.0	47.2	43.5	38.3	27.2	43.9	41.9	40.7	29.9	35.6	35.8	51.2	49.0	39.6
mContriever	43.1	52.5	50.1	36.4	41.8	21.5	60.2	31.4	28.6	39.2	42.4	48.3	39.1	56.0	52.8	51.7	41.0	40.8	41.5
mE5 _{large}	66.6	76.0	75.9	52.9	52.9	59.0	77.8	54.5	62.0	52.9	70.6	66.5	67.4	74.9	84.6	80.2	56.0	56.4	78.3
E5 _{mistral-7b}	63.4	73.3	70.3	57.3	52.2	52.1	74.7	55.2	52.1	52.7	66.8	61.8	67.7	68.4	73.9	74.0	54.0	54.1	79.7
OpenAI-3	54.9	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
BGE-M3 (Dense)	69.2	78.4	80.0	56.9	56.1	60.9	78.6	58.3	59.5	56.1	72.8	69.9	70.1	78.7	86.2	82.6	62.7	56.7	81.8
mGTE-TRM (Dense)	62.1	71.4	72.7	54.1	51.4	51.2	73.5	53.9	51.6	50.3	65.8	62.7	63.2	69.9	83.0	74.0	60.8	49.7	58.3
bge-multilingual-gemma2	74.1	81.0	82.3	64.5	64.2	64.0	81.2	64.2	68.2	61.5	79.1	69.7	77.0	81.9	88.1	84.6	68.0	63.5	90.3

Table 10: Multi-lingual retrieval performance on the MIRACL (Zhang et al., 2023) dev set (measured by nDCG@10).

Model	Avg.	ar	bn	en	es	fa	fi	fr	hi	id	ja	ko	ru	sw	te	th	zh	de	yo
BM25	67.3	78.7	90.0	63.6	25.4	68.1	81.2	50.2	73.8	71.8	73.6	70.1	56.4	69.9	73.3	87.5	55.1	42.8	80.1
mDPR	79.0	84.1	81.9	76.8	86.4	89.8	78.8	91.5	77.6	57.3	82.5	73.7	79.7	61.6	76.2	67.8	94.4	89.8	71.5
mContriever	84.9	92.5	92.1	79.7	84.1	65.4	95.3	82.4	64.6	80.2	87.8	87.5	85.0	91.1	96.1	93.6	90.3	84.1	77.0
mE5 _{large}	94.1	97.3	98.2	87.6	89.1	92.9	98.1	90.6	93.9	87.9	97.1	93.4	95.5	96.7	99.2	98.9	93.3	90.7	93.1
E5 _{mistral-7b}	92.7	96.0	96.0	90.2	87.5	88.0	96.7	92.8	89.9	88.4	95.1	89.4	95.0	95.5	95.1	96.5	90.1	88.7	97.9
BGE-M3 (Dense)	95.5	97.6	98.7	90.7	91.1	94.0	97.9	93.8	94.4	90.5	97.5	95.5	95.9	97.2	99.4	99.1	96.9	90.9	98.7
bge-multilingual-gemma2	97.2	99.0	98.9	95.4	94.5	95.0	98.5	96.2	96.5	95.3	98.9	95.4	98.0	98.0	99.7	99.6	97.2	94.1	100.0

Table 11: Multi-lingual retrieval performance on the MIRACL (Zhang et al., 2023) dev set (measured by Recall@100).

C.2 MAIN RESULTS

MIRACL. Following BGE-M3 (Chen et al., 2024), we evaluate the multilingual retrieval performance with MIRACL (Zhang et al., 2023). We cite most of the results reported in the Table 1 of BGE-M3’s paper. It should be noted that the results of BM25 are lower than the results reported in MIRACL’s paper, as the BM25 tested in BGE-M3’s paper used the same tokenizer with BGE-M3. We also include another recent work mGTE (Zhang et al., 2024) as one of the baseline models.

As shown in Table 10, our model *bge-multilingual-gemma2* achieves the state-of-the-art (SOTA) performance in all 18 languages. The overall performance of *bge-multilingual-gemma2* is 74.1, which is far ahead of the performance of the previous best model BGE-M3 (Dense), indicating the excellent multilingual retrieval capability of *bge-multilingual-gemma2*. The results of Recall@100 are available in Table 11.

FR-MTEB & PL-MTEB & C-MTEB. We further evaluate our model on FR-MTEB (Ciancone et al., 2024), PL-MTEB (Poświata et al., 2024) and C-MTEB (Xiao et al., 2024) benchmarks. FR-MTEB consists of 26 datasets in 6 different tasks, PL-MTEB consists of 26 datasets in 5 different tasks, and C-MTEB consists of 35 datasets in 6 different tasks. We use the API provided by MTEB ² to perform evaluation.

The results shown in Table 12, Table 13 and Table 14 are all available in the MTEB leaderboard ³. We can observe that *bge-multilingual-gemma2* leads to new SOTA performances on both FR-MTEB and PL-MTEB benchmarks, and especially achieves very excellent results in the retrieval tasks (Retr.). On the C-MTEB benchmark, *bge-multilingual-gemma2* surpasses most of the baseline models, such as e5-mistral-7b-instruct, bge-large-zh-v1.5, etc. However, its overall performance on C-MTEB benchmark is still slightly worse than gte-Qwen2-7B-instruct, which can be attributed to Gemma-2’s Chinese proficiency being worse than that of Qwen2.

MTEB. The evaluation results of *bge-multilingual-gemma2* on the MTEB benchmark are available in Table 1. The detailed results for each dataset are available in Table 8. We can also observe that *bge-multilingual-gemma2* achieves good performance on the MTEB benchmark.

²<https://github.com/embeddings-benchmark/mteb>

³<https://huggingface.co/spaces/mteb/leaderboard>

Task # of datasets →	Retr. 5	Rerank. 2	Clust. 7	PairClass. 2	Class. 6	STS 3	Summ. 1	Avg. 26
mistral-embed	46.81	80.46	44.74	77.32	68.61	79.56	31.47	59.41
gte-multilingual-base	52.97	76.47	41.66	79.46	68.72	81.36	29.74	59.79
voyage-multilingual-2	54.56	82.59	46.57	78.66	68.56	80.13	29.96	61.65
gte-Qwen2-1.5B-instruct	52.56	83.76	55.01	86.88	78.02	81.26	30.5	66.6
gte-Qwen2-7B-instruct	55.65	78.7	55.56	90.43	81.76	82.31	31.45	68.25
bge-multilingual-gemma2	63.47	85.22	56.48	85.07	81.62	82.59	31.26	70.08

Table 12: Results on the FR-MTEB (Ciancone et al., 2024) benchmark (26 datasets in the French subset). Please refer to Table 17 for the scores of bge-multilingual-gemma2 per dataset.

Task # of datasets →	Retr. 11	Clust. 1	PairClass. 4	Class. 7	STS 3	Avg. 26
gte-multilingual-base	46.40	33.67	85.45	60.15	68.92	58.22
multilingual-e5-large	48.98	33.88	85.50	63.82	66.91	60.08
mmlw-roberta-large	52.71	31.16	89.13	66.39	70.59	63.23
gte-Qwen2-1.5B-instruct	51.88	44.59	84.87	72.29	68.12	64.04
gte-Qwen2-7B-instruct	54.69	51.36	88.48	77.84	70.86	67.86
bge-multilingual-gemma2	59.41	50.29	89.62	77.99	70.64	70.00

Table 13: Results on the PL-MTEB (Poświata et al., 2024) benchmark (26 datasets in the Polish subset). Please refer to Table 17 for the scores of bge-multilingual-gemma2 per dataset.

Task # of datasets →	Retr. 8	Rerank. 4	Clust. 4	PairClass. 2	Class. 9	STS 8	Avg. 35
multilingual-e5-large	63.66	56.00	48.23	69.89	67.34	48.29	58.81
e5-mistral-7b-instruct	61.75	61.86	52.30	72.19	70.17	50.22	60.81
gte-multilingual-base	71.95	68.17	47.48	78.34	64.27	52.73	62.72
bge-large-zh-v1.5	70.46	65.84	48.99	81.6	69.13	56.25	64.53
gte-Qwen2-1.5B-instruct	71.86	68.21	54.61	86.91	71.12	60.96	67.65
gte-Qwen2-7B-instruct	76.03	68.92	66.06	87.48	75.09	65.33	72.05
bge-multilingual-gemma2	73.73	68.28	59.3	86.67	74.11	56.87	68.44

Table 14: Results on the C-MTEB (Xiao et al., 2024) benchmark (35 datasets in the Chinese subset). Please refer to Table 17 for the scores of bge-multilingual-gemma2 per dataset.

Domain # of datasets →	wiki 2	web 2	news 2	healthcare 2	law 1	finance 2	arxiv 1	msmarco 1	Avg. 13
bge-m3 (Dense)	61.42	48.86	44.40	45.74	26.68	41.85	40.76	54.40	46.65
multilingual-e5-large	57.16	42.91	41.61	42.18	19.66	37.38	36.93	54.44	42.58
e5-mistral-7b-instruct	58.82	45.18	42.08	46.06	19.32	40.45	44.78	59.03	45.26
gte-Qwen2-1.5B-instruct	55.04	42.95	37.30	44.50	11.95	40.24	32.06	49.74	41.06
gte-Qwen2-7B-instruct	64.95	51.59	48.55	46.51	22.31	42.42	40.27	58.39	48.38
bge-multilingual-gemma2	65.50	51.81	47.46	44.68	22.58	40.45	23.28	63.14	46.83

Table 15: QA (en & zh, nDCG@10) performance on AIR-Bench 24.04.

AIR-Bench. For the QA task in AIR-Bench, we perform evaluation on all of the 13 datasets in 24.04 version, which consists of 8 English datasets and 5 Chinese datasets. For the Long-Doc task in AIR-Bench, we perform evaluation on all of the 15 datasets in 24.04 version, which are all English datasets. As shown in Table 15 and Table 3, bge-multilingual-gemma2 also achieves excellent performance in the out-of-distribution (OOD) evaluation on AIR-Bench, which indicates that our model has excellent generalization ability.

Task Name	Instruction Template
C-MTEB	
CLSClusteringS2S	Identify the main category of scholar papers based on the titles.
CLSClusteringP2P	Identify the main category of scholar papers based on the titles and abstracts.
ThuNewsClusteringS2S	Identify the topic or theme of the given news articles based on the titles.
ThuNewsClusteringP2P	Identify the topic or theme of the given news articles based on the titles and contents.
T2Reranking	Given a Chinese search query, retrieve web passages that answer the question.
MMarcoReranking	Given a Chinese search query, retrieve web passages that answer the question.
CMedQAv1	Given a Chinese community medical question, retrieve replies that best answer the question.
CMedQAv2	Given a Chinese community medical question, retrieve replies that best answer the question.
Ocnli	Retrieve semantically similar text.
Cmnli	Retrieve semantically similar text.
T2Retrieval	Given a Chinese search query, retrieve web passages that answer the question.
MMarcoRetrieval	Given a web search query, retrieve relevant passages that answer the query.
DuRetrieval	Given a Chinese search query, retrieve web passages that answer the question.
CovidRetrieval	Given a question on COVID-19, retrieve news articles that answer the question.
CmedqaRetrieval	Given a Chinese community medical question, retrieve replies that best answer the question.
EcomRetrieval	Given a user query from an e-commerce website, retrieve description sentences of relevant products.
MedicalRetrieval	Given a medical question, retrieve user replies that best answer the question.
VideoRetrieval	Given a video search query, retrieve the titles of relevant videos.
PL-MTEB	
CBD	Classify the sentiment of polish tweet reviews.
PolEmo2.0-IN	Classify the sentiment of in-domain (medicine and hotels) online reviews.
PolEmo2.0-OUT	Classify the sentiment of out-of-domain (products and school) online reviews.
AllegroReviews	Classify the sentiment of reviews from e-commerce marketplace Allegro.
PAC	Classify the sentence into one of the two types: "BEZPIECZNE.POSTANOWIENIE.UMOWNE" and "KLAUZULA.ABUZYWNA".
8TagsClustering	Identify of headlines from social media posts in Polish into 8 categories: film, history, food, medicine, motorization, work, sport and technology.
SICK-E-PL	Retrieve semantically similar text.
PPC	Retrieve semantically similar text.
CDSC-E	Retrieve semantically similar text.
PSC	Retrieve semantically similar text.
ArguAna-PL	Given a claim, find documents that refute the claim.
DBPedia-PL	Given a query, retrieve relevant entity descriptions from DBPedia.
FiQA-PL	Given a financial question, retrieve user replies that best answer the question.
HotpotQA-PL	Given a multi-hop question, retrieve documents that can help answer the question.
MSMARCO-PL	Given a web search query, retrieve relevant passages that answer the query.
NFCorpus-PL	Given a question, retrieve relevant documents that best answer the question.
NQ-PL	Given a question, retrieve Wikipedia passages that answer the question.
Quora-PL	Given a question, retrieve questions that are semantically equivalent to the given question.
SCIDOCS-PL	Given a scientific paper title, retrieve paper abstracts that are cited by the given paper.
SciFact-PL	Given a scientific claim, retrieve documents that support or refute the claim.
Touche2020	Given a question, retrieve detailed and persuasive arguments that answer the question.
TRECCOVID-PL	Given a query, retrieve documents that answer the query.
FR-MTEB	
MasakhaNEWSClassification	Classify the given news article into one of the seven topic categories: politics, sports, health, business, entertainment, technology, and religion.
AlloProfClusteringP2P	Identify the main category of Allo Prof document based on the titles and descriptions.
AlloProfClusteringS2S	Identify the main category of Allo Prof document based on the titles.
HALClusteringS2S	Identify the main category of academic passage based on the titles and contents.
MasakhaNEWSClusteringP2P	Identify the topic or theme of the given news articles based on the titles and contents.
MasakhaNEWSClusteringS2S	Identify the topic or theme of the given news articles based on the titles.
MLSUMClusteringP2P	Identify the topic or theme of the given articles based on the titles and contents.
MLSUMClusteringS2S	Identify the topic or theme of the given articles based on the titles.
AlloprofReranking	Given a question, retrieve passages that answer the question.
OpusparcusPC	Retrieve semantically similar text.
PawsXPairClassification	Retrieve semantically similar text.
SyntecReranking	Given a question, retrieve passages that answer the question.
AlloprofRetrieval	Given a question, retrieve passages that answer the question.
BSARDRetrieval	Given a question, retrieve passages that answer the question.
SyntecRetrieval	Given a question, retrieve passages that answer the question.
XPQARetrieval	Given a question, retrieve passages that answer the question.
MintakaRetrieval	Given a question, retrieve passages that answer the question.

Table 16: The additional instruction we used on the C-MTEB, PL-MTEB and FR-MTEB benchmarks. These instructions are adopted from gte-Qwen2-7B-instruct (Li et al., 2023). To ensure sentence completeness, we add a period at the end.

Dataset	Result	Dataset	Result
FR-MTEB		PL-MTEB	
AlloprofRetrieval	58.50	ArguAna-PL	59.71
BSARDRetrieval	28.52	DBPedia-PL	43.19
MintakaRetrieval (fr)	62.53	FiQA-PL	46.12
SyntecRetrieval	90.37	HotpotQA-PL	77.03
XPQARetrieval (fr)	77.42	MSMARCO-PL	72.69
AlloprofReranking	78.62	NFCorpus-PL	36.72
SyntecReranking	91.83	NQ-PL	56.85
AlloProfClusteringP2P	71.20	Quora-PL	84.47
AlloProfClusteringS2S	59.64	SCIDOCs-PL	19.53
HALClusteringS2S	28.19	SciFact-PL	74.43
MLSUMClusteringP2P (fr)	47.75	TRECCOVID-PL	82.75
MLSUMClusteringS2S (fr)	47.46	8TagsClustering	50.29
MasakhaNEWSClusteringP2P (fra)	73.86	CDSC-E	78.23
MasakhaNEWSClusteringS2S (fra)	67.24	PPC	95.43
OpusparcusPC (fr)	100.00	PSC	99.24
PawsXPairClassification (fr)	70.14	SICK-E-PL	85.58
AmazonReviewsClassification (fr)	55.19	AllegroReviews	65.00
MasakhaNEWSClassification (fra)	82.49	CBD	84.13
MassiveIntentClassification (fr)	79.60	MassiveIntentClassification (pl)	79.41
MassiveScenarioClassification (fr)	82.18	MassiveScenarioClassification (pl)	81.93
MTOPDomainClassification (fr)	97.20	PAC	67.24
MTOPIntentClassification (fr)	93.07	PolEmo2.0-IN	90.42
STS22 (fr)	83.28	PolEmo2.0-OUT	77.77
STSBenchmarkMultilingualSTS (fr)	85.09	CDSC-R	90.97
SICKFr	79.39	SICK-R-PL	78.16
SummEvalFr	31.26	STS22 (pl)	42.79
FR-MTEB Average (23)	70.08	PL-MTEB Average (23)	70.00
C-MTEB			
CmedqaRetrieval	42.21	AmazonReviewsClassification (zh)	54.34
CovidRetrieval	77.46	IFlyTek	49.94
DuRetrieval	90.46	JDReview	88.91
EcomRetrieval	69.3	MassiveIntentClassification (zh-CN)	78.19
MedicalRetrieval	62.02	MassiveScenarioClassification (zh-CN)	82.58
MMarcoRetrieval	84.7	MultilingualSentiment	78.91
T2Retrieval	86.26	OnlineShopping	94.59
VideoRetrieval	77.4	TNews	50.26
CMedQAv1	84.62	Waimai	89.26
CMedQAv2	85.60	AFQMC	47.17
MMarcoReranking	35.43	ATEC	50.75
T2Reranking	67.48	BQ	62.02
CLSClusteringP2P	54.65	LCQMC	75.95
CLSClusteringS2S	63.68	PAWSX	30.57
ThuNewsClusteringP2P	64.32	QBQTC	38.98
ThuNewsClusteringS2S	54.57	STS22 (zh)	68.68
Cmnli	90.13	STSB	80.87
Ocnli	83.21	C-MTEB Average (35)	68.44

Table 17: Results of **bge-multilingual-gemma2** for each dataset in the FR-MTEB, PL-MTEB and C-MTEB benchmarks.

D LIGHTWEIGHT RE-RANKER

We have also introduced a lightweight version of the reranker, which incorporates both depth and width compression techniques. Specifically, depth compression is implemented on a layerwise method, allowing for the selective adjustment of the number of layers according to the desired output. Regarding width compression, it is configured to execute token compression at predetermined layers, whereby n tokens are merged into a single token.

For the input template, we use the following format:

$$A: \{\text{query}\} \quad B: \{\text{passage}\} \quad \{\text{prompt}\} \quad (7)$$

where the prompt inquires about the relationship between A and B, e.g., *Predict whether passage B contains an answer to query A*. And we use the logits of *Yes* as our reranking score.

Considering the depth compression generates output scores at each layer, we extract the linear layer connected to the logits for the *Yes* prediction in the language model head. This extracted linear layer is then appended to each layer, allowing every layer to compute a reranking score.

D.1 SETUP

LLM. Our objective is to develop a multilingual version of the lightweight reranker. Considering the extensive vocabulary necessitated by multilingual support, we employ Gemma-2-9b (Team et al., 2024) as the backbone for our reranker.

Dataset. We evaluate the performance of our reranker *bge-reranker-v2.5-gemma-lightweight* on BEIR (Thakur et al., 2021) and MIRACL (Zhang et al., 2023). The BEIR benchmark encompasses a variety of text retrieval tasks across multiple domains, while MIRACL serves as a significant dataset for multilingual evaluation, featuring 18 distinct languages.

Training Data. To enhance the multilingual capabilities and retrieval performance of the Reranker, we utilize the BGE-M3 dataset (Chen et al., 2024), along with Arguana, HotpotQA, and FEVER, for the training process.

Training Detail. The reranker is trained using contrastive loss. Furthermore, LoRA is employed for fine-tuning, where the LoRA rank is set to 64 and the LoRA alpha is set to 32, accompanied by a learning rate of $1e-4$. During the training process, a batch size of 128 is utilized, and 15 hard negatives are assigned to each query. At the same time, the training of the reranker employs self-distillation, wherein the final layer serves as the teacher for preceding layers. Throughout this training process, KL divergence loss is utilized. During training, we randomly select a width compression strategy and train all depth compression strategies. Regarding depth compression, we support outputs from 8 to 42 layers. Regarding width compression, we support compression ratios of 1, 2, 4, and 8, and support width compression at 8, 16, 24, 32, and 40 layers. During the training process, we utilized four types of prompts: query to passage, query to query, passage to passage, and argument to counter-argument. The specific application of these prompts was dependent on the type of dataset used, as shown in Table 18.

Evaluation. On the BEIR benchmark, we rerank the top-100 retrieval results of *bge-large-en-v1.5* and *E5-mistral-7b-instruct*. On the MIRACL dataset, we rerank the top-100 retrieval results of *bge-m3* (dense). The instructions for evaluation are shown in Table 19.

Task Type	Instruction Template
query to passage	Predict whether passage B contains an answer to query A.
query to query	Predict whether queries A and B are asking the same thing.
passage to passage	Predict whether passages A and B have the same meaning.
argument to counter-argument	Predict whether argument A and counterargument B express contradictory opinions.

Table 18: The training instructions we used for reranker.

Task Name	Instruction Template
ArguAna	Predict whether argument A and counterargument B express contradictory opinions.
ClimateFEVER	Predict whether passage B contains an answer to query A.
CQADupstack	Predict whether queries A and B are asking the same thing.
DBPedia	Predict whether passage B contains an answer to query A.
FEVER	Predict whether passage B contains an answer to query A.
FiQA2018	Predict whether passage B contains an answer to query A.
HotpotQA	Predict whether passage B contains an answer to query A.
MSMARCO	Predict whether passage B contains an answer to query A.
NFCorpus	Predict whether passage B contains an answer to query A.
Natural Question	Predict whether passage B contains an answer to query A.
QuoraRetrieval	Predict whether queries A and B are asking the same thing.
SCIDOCS	Predict whether passage B contains an answer to query A.
SciFact	Predict whether passage B contains an answer to query A.
Touche2020	Predict whether passage B contains an answer to query A.
TREC-COVID	Predict whether passage B contains an answer to query A.
MIRACL	Predict whether passage B contains an answer to query A.

Table 19: The instructions we used for the BEIR benchmark and MIRACL dataset for reranker.

BEIR	bge-large-en-v1.5	bge-reranker-v2-m3	jina-reranker-v2-base-multilingual	bge-reranker-v2-gemma	bge-reranker-v2.5-gemma-lightweight	bge-reranker-v2.5-gemma-lightweight
Save Flops	-	-	-	-	60%	0
ArguAna	63.54	37.70	52.23	78.68	86.04	86.16
ClimateFEVER	36.49	37.99	34.65	39.07	48.41	48.48
CQA	42.23	38.24	40.21	45.85	49.18	48.9
DBPedia	44.16	48.15	49.31	49.92	51.98	52.11
FEVER	87.17	90.15	92.44	90.15	94.71	94.69
FiQA2018	44.97	49.32	45.88	49.32	60.48	60.95
HotpotQA	74.11	84.51	81.81	86.15	87.84	87.89
MSMARCO	42.48	47.79	47.83	48.07	47.23	47.26
NFCorpus	38.12	34.85	37.73	39.73	41.40	41.64
NQ	55.04	69.37	67.35	72.60	75.37	75.58
QuoraRetrieval	89.06	89.13	87.81	90.37	91.25	91.18
SCIDOCS	22.62	18.25	20.21	21.65	23.71	23.87
SciFact	74.64	73.08	76.93	77.22	80.5	80.38
Touche2020	25.08	35.68	32.45	35.68	30.64	31.09
TRECCOVID	74.89	83.39	80.89	85.51	84.26	84.85
Mean	54.31	55.36	56.52	60.71	63.10	63.67

Table 20: The performance of various rerankers on BEIR benchmark (based on bge-large-en-v1.5).

D.2 MAIN RESULTS

BEIR. We rerank the retrieval results from the BEIR dataset using two models, bge-large-en-v1.5 and E5-Mistral-7b-Instruct, and we rerank top-100 retrieval results from these models. We conduct both a comprehensive evaluation and a lightweight evaluation. In the lightweight evaluation, we select a compression ratio of 2, a width compression factor of 8, and a depth of the output layer set to 25. This configuration results in a 60% FLOPs.

Tables 20 and 21 present the evaluation results for the BEIR benchmark. It indicates that bge-reranker-v2.5-gemma2-lightweight records exceptional performance in enhancing both bge-large-en-v1.5 and E5-Mistral-7b-Instruct retrieval outcomes. Furthermore, there exists a positive correlation between the initial retrieval quality and the subsequent reranking performance, when reranking the retrieval results from E5-Mistral-Instruct, our reranker achieves improved performance. Addi-

BEIR	E5-mistral-7b-instruct	bge-reranker-v2-gemma	bge-reranker-v2.5-gemma-lightweight	bge-reranker-v2.5-gemma-lightweight
Save FLOPs	-	-	60%	0
ArguAna	61.80	79.05	86.02	86.58
ClimateFEVER	38.37	37.66	47.27	47.13
CQA	42.97	46.16	49.06	49.53
DBPedia	48.84	50.77	52.45	52.87
FEVER	87.82	91.36	94.85	95.19
FiQA2018	56.58	50.96	58.81	61.19
HotpotQA	75.72	86.99	88.49	88.82
MSMARCO	43.06	48.35	47.65	47.40
NFCorpus	38.58	39.25	42.28	42.17
NQ	63.56	73.44	75.00	76.28
QuoraRetrieval	89.59	90.44	91.09	91.18
SCIDOCS	16.30	20.77	22.20	22.69
SciFact	76.26	77.78	79.94	80.98
Touche2020	26.24	35.79	28.69	31.17
TRECCOVID	87.07	88.13	86.61	87.36
Mean	56.85	61.13	63.36	64.04

Table 21: The performance of various rerankers on BEIR benchmark (based on E5-mistral-7b-instruct).

tionally, the implementation of the lightweight model variant results in only a marginal decline in performance while achieving a significant reduction in FLOPs.

Language	bge-m3 (Dense)	bge-reranker-v2-gemma	bge-reranker-v2-m3	bge-reranker-v2-gemma	bge-reranker-v2.5-gemma-lightweight	bge-reranker-v2.5-gemma-lightweight
FLOPS	-	-	-	-	60%	0
ar	78.4	73.4	81.7	82.3	82.5	82.8
bn	80.0	81.9	84.6	85.0	87.8	87.6
en	56.9	58.9	63.5	66.6	68.6	69.3
es	56.1	58.6	64.4	65.3	67.6	67.8
fa	60.9	60.5	65.7	65.5	67.5	67.4
fi	78.6	77.2	82.4	82.6	82.8	83.3
fr	58.3	56.1	63.7	65.4	68.5	68.5
hi	59.5	62.7	68.5	69.4	71.4	71.3
id	56.1	59.6	62.7	61.2	63.8	63.8
ja	72.8	72.7	80.0	79.7	82.8	83.6
ko	69.9	74.0	73.8	75.1	75.9	75.7
ru	70.1	67.1	76.9	78.3	79.8	80.1
sw	78.7	78.1	82.3	81.8	84.8	85.1
te	86.2	85.8	89.4	89.6	90.8	90.8
th	82.6	81.2	85.3	86.1	88.1	88.7
zh	62.7	63.0	65.2	66.8	69.9	69.9
de	56.7	58.2	62.7	64.0	65.8	65.6
yo	81.8	84.2	87.4	85.9	89.6	89.8
Mean (18)	69.2	69.6	74.4	75.0	77.1	77.3

Table 22: Comparison of MIRACL dev nDCG@10 scores across various rerankers (based on bge-m3 (Dense)).

MIRACL. We further evaluate the multilingual capabilities of the reranker using the MIRACL dataset, with the results presented in Table 22. The reranking is conducted based on the top 100 retrieval results obtained from the bge-m3 (dense) model. The reranker demonstrates a significant improvement in retrieval accuracy across each dataset and outperforms other multilingual rerankers. Notably, compared to monolingual (English) retrieval, the multilingual retrieval experienced minimal negative effects from model lightweighting, essentially maintaining the original performance of the model.